

LEZIONE 21/04 pt.1

MACHINE LEARNING for vocal analysis.

Insight into machine learning approaches for health monitoring.

The goal is projecting a system capable of monitoring our status of health.

The tools are non intended to provide a definitive diagnosis, but to **help clinicians triage potential cases.**

Important to understand 10/20 years earlier than manifestation of symptoms if patient is starting to develop the issue.

POSSIBLE APPLICATIONS

Automatic vocal evaluation finds application to the analysis of any pathology that directly or indirectly affects the vocal apparatus.

When a person is **depressed**, the voice changes. They speak at a constant amplitude and loudness, not having peaks. We can classify in sad/happy people.

We can detect **respiratory diseases**, because covid, for example, affects vocal apparatus.

We can analyse **sleep disorder**, important markers of neurodegeneration.

It is the same for **eating disorders**. In obesity, the amount of body-fat influences the capability to produce the sounds.

PARKINSON'S DESEASE identified by analysing acoustic features of each sustained vowel phonation.

ALZHEIMER'S DESEASE: **shorter words**, **small vocabulary**, more **fragmented sentences**.

AUTISM SPECTRUM: analysing **babbling** of infants.

POST TRAUMATIC STRESS: people change their emotion, flatter and more monotonous speak.

CORONARY ANGIOGRAM.

Speaking requires the coordination of numerous anatomical structures and system.

Lungs: produce a flow that passes through the larynges and arrive to vocal cords. Can be close together or apart. If together, they produce sounds that are shaped by tongue, lips and nasal structure.

If the cords are a part, do not vibrate, the sound produced is not periodic and called unvoiced sounds (consonant).

All this is coordinated from brain.

A disease that affects any one of these systems might leave **diagnostic clues** in a person's speech.

Alterations to the vocal signal can be measured and input into machine learning algorithms (ML), implementing beneficial tools to support clinical practice.

Researchers and clinician could simply feed hundreds or thousands of voice samples into a computer to search for features that distinguish people with various medical conditions.

ADVANTAGES:

- *EASY COLLECTION*
- *USEFUL INFORMATION*
- *OBJECTIVE MEASURES*
- *LOW-COST*
- *REMOTE MONITORING.*

The **aim of the project** is to develop a web /smartphone app that will be able to enhance frequent and remote monitoring of the patient's health condition.

The app must include:

- Tailored exercise
- Personalized application
- Objective measures
- Monitoring
- Assistance
- Education.

CASE STUDY: PARKINSON'S DISEASE.

Degeneration of dopamine neurons, especially in the substantia nigra pars compacta region of the midbrain.

Progressive disability with impact on ADLs (activities of daily living).

Motor/ Non motor symptoms.

It affects 1% of over 60 population, 9 million people. **90% of them presents speech alteration, that comes out 10 years before the manifestation of the disease.**

We want to:

COLLECT DATA -> EXTRACT INFORMATION -> ANALYSE DATA.

The **extract information** is made of:

- Pre-Processing (silence removal, de-noising, normalization, regions identification, framing)
- Feature extraction (high-level/low-level features): **PHONATION, ARTICULAT, PROSODY.**
- Feature selection

PHONATION	Mainly associated to the ability of producing air in the lungs to make vocal folds vibrate to produce vocal sounds
ARTICULATION	Refers to the use of special districts of the vocal tract to produce syllables and words.
PROSODY	Is related to amplitude, loudness and rhythmic organization of the speech.

SPEECH ALTERATION IN PARKINSON'S DISEASE PATIENS.

Lack of controls of the vocal folds and articulators
Inability to perform fine, rapid, and synergistic movements
Reduced amplitude and loudness of speech
Impaired fluency with repetitive phoneme

The **analyse data** is made of:

- Investigate possible correlations and trends
- Perform classification
- Perform regression.

The occlusive consonant is the most affected in Italian language. (T, P). Are produced by a rapid movement that firstly blocks air behind the vocal cords and releases their positions and makes air flows between them.

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